Tutorial - 5

24/2/26

SVM

- class (spam = +1, Not spam = -1)

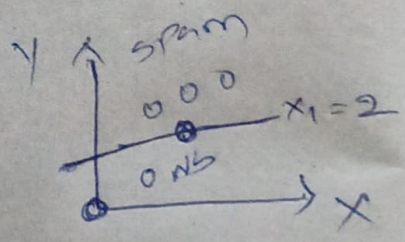
(Bias: w_0)

Suspicious words	Links	Attachment	class (y)
6	3	1	+1
1	0	0	-1
4	2	1	+1
0	0	0	-1
5	2	1	+1
(w_1)	(w_2)	(w_3)	

weighted sum : $Z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$

	x_1	x_2	x_3
Spam →	(4, 5, 6)	(2, 3)	(1)
Non-spam →	(0, 1)	(0)	(0)

Initialize : w_2 & $w_3 = 0$



spam

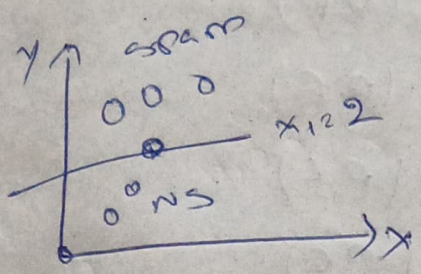
$f(x) = 0$

non-spam

Decision Boundary

If $f(x) = 0 \Rightarrow w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$

$\Rightarrow w_0 + w_1 x_1 = 0$



now; $w_1 = 1$

$w_0 + w_1 x_1 = 0$

$w_0 + (1.2) = 0$

$\boxed{w_0 = -2}$

$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 = -2$

calculate
or
check

$y_i(x_1 w_1 + x_2 w_2 + x_3 w_3 + w_0) \geq 1 \rightarrow$ perfect classification

$\leq 1 \rightarrow$ classification fail

① $= 1(6(1) + 3(0) + 1(0) + (-2))$

$= 1(6-2)$

$= 4 > 1$

check

$y_i(x_1 - 2)$

(2)

$-(1-2)$

$= 1$

(3)

$1(4-2)$

$= 2$

(4)

$-1(0-2)$

$= 2$

$\hat{y}$
≥ 1 (P.F)
≥ 1 (P.F)
≥ 1 (P.F)
≥ 1 (P.F)
≥ 1 (P.F)

cost or loss f^n

$$\Rightarrow L = \frac{1}{2}\|w\|^2 - \sum_{i} c_i \left[y_i (w x_i + b) - 1\right]$$

classification error hyperparameter

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} c_i \text{ error}_i$$

$y - \hat{y}$

Update weights

If P.F $\Rightarrow y_i(w_1 x_1 + w_2 x_2 + \dots) \geq 1$ $\longrightarrow L = \frac{1}{2} w^2$

$c \to 0$

$$w^{new} = w^{old} - \eta \frac{\partial L}{\partial w}$$

$$= w - \eta w$$

$$\boxed{w_1^{new} = w_1^{old} - \eta w_1^{old}}$$

$$w_0^{new} = w_0^{old} - \eta w_0^{old}$$

$$\boxed{\eta = 0.1}$$

$$w_0^{new} = 1 - (0.1 \times 1)$$

$$= 1 - 0.1$$

$$= 0.9$$

$$\text{Margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^3 + \dots}}$$

$\Rightarrow$ If N.P.F $\Rightarrow y_i(w_1 x_1 + w_2 x + \dots) < 1$

Take C

$$w^{new} = w - \eta \frac{\partial L}{\partial w}$$

$$= w - \eta \left[w - \sum_i c_i y_i x_i \right]$$

$$= (1-\eta) w + \eta \sum c_i y_i x_i$$

$$\sum c_i y_i x_i = c \sum_i y_i x_i$$

$$c = 1 \Rightarrow y_1 x_1 + y_2 x_{12} + y_3 x_{13} + y_4 y_{14} + y_5 y_{15})$$

SVM

weighted sum: $f(x) = z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$

Goal: check: $y_i [f(x)] \geq 1 \rightarrow$ perfect classification

$< 1 \Rightarrow$ classification fails

Initialize $w_1 = 1$

$w_2 = 1$

$w_3 = 1$

$b = 0$

1st set of Inputs: $y_1 (f(x)) = -10 < 1$

$$w_j^{new} = w_j^{old} - \eta \frac{\partial L}{\partial w}$$

$$w_j^{new} = w_j^{old} - \eta \left[w = \sum_i c_i y_i x_{ij} \right]$$

$$w_1^{new} = w_1^{old} - \eta \left[w_1^{old} - \sum y_i x_i \right] \quad \boxed{c = 1}$$

$$w_1^{new} = w_j^{old} - \eta \, x_{y0} \frac{\partial L}{\partial w}$$

$$w_1^{new} = 1 - 1 \left[1 - (-1 \times 6) \right] = -6$$

(x_1)	(x_2)	(x_3)	(y)
6	3	1	+1
1	0	0	−1
4	2	1	+1
0	0	0	−1
5	2	1	+1

$w_2^{new} = 1 - 1[1 - (-1 \times 3)] = -3$

$w_3^{new} = 1 - 1[1 - (-1 \times 1)] = -1$

$b^{new} = b^{old} - \eta \frac{\partial L}{\partial b}$

$= b^{old} - \eta [\Sigma c_i y_i]$

$= 0 - [(-1)] = 1$

$\eta = 1$
$b = 0$
$c = 1$
$y = 1$

$\boxed{\text{for perfect } c = 0}$

2nd set of inputs

$w_1 = -6$
$w_2 = -3$
$w_3 = -1$
$b = 1$

$y_i \times f(x_i) = -1(-6+1) = 5 > 1$

no need to add error term

If Right Prediction : $L = \frac{1}{2}\|w\|^2$

$\frac{\partial L}{\partial w} = w$

$w_j^{new} = w_j^{old} - \eta \frac{\partial L}{\partial w}$

$w_j^{new} = w_j^{old} - \eta w_j^{old}$

$w_1^{new} = -6 - (1 - 6) = 0$
$w_2^{new} = -3 - (1 - 3) = 0$
$w_3^{new} = -1 - (1 - 1) = 0$

so take η as 0-1

Q) calculate the margin:

$$\text{margin} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$b^{new} = b^{old} - \eta \frac{\partial L}{\partial b}$$

$$= b^{old} - \eta (\sum c_i y_i)$$

$$= 1 - [0]$$

$$b^{new} = 1$$